

ganit

data speaks

PHARMA | INDIA & GLOBAL

Cipla – Enterprise BI & Data Engineering Platform

Unifying Commercial, Manufacturing & Regulatory Analytics with a Scalable Azure Data Platform

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Ganit Confidential

Cipla

Enterprise BI & Data Engineering

40%

Faster Report Delivery

15+

Data Sources Unified

99.9%

Pipeline Uptime

60%

Cost Reduction

Unifying Commercial, Manufacturing & Regulatory Analytics for a Global Pharma Leader

Problem Statement	Solution Overview	Business Value
▶ Siloed commercial & manufacturing data ▶ Manual Excel-based compliance reports ▶ No real-time supply chain visibility ▶ Fragmented CRM & ERP systems	▶ Azure Data Lake + Synapse DWH ▶ ADF + Databricks pipelines ▶ Power BI Premium dashboards ▶ Purview governance & data catalog	▶ Unified view across 80+ markets ▶ Near-real-time regulatory reporting ▶ Automated pharma compliance alerts ▶ Self-serve analytics for 500+ users

Azure Services Used: ADLS Gen2 | Azure Data Factory | Databricks | Synapse Analytics | Power BI Premium | Purview | Key Vault | Azure AD

Outcome

Regulatory Compliance

Automated schedule M, WHO-GMP, and pharmacovigilance reporting — eliminating manual reconciliation and reducing submission errors by 88% across all regulatory jurisdictions.

Cost Optimisation

Replaced 4 legacy on-premise data warehouses with a unified Azure platform, achieving ~60% reduction in infrastructure costs while enabling elastic scaling for batch reporting peaks.

Risk & Resilience

Built a multi-zone ADLS Gen2 data lake (Raw/Cleansed/Curated) with Purview lineage — handling 12+ TB of historical pharma data and supporting 500+ concurrent Power BI users globally.

Business Growth

Enabled real-time commercial sales, inventory, and demand dashboards in Power BI — giving supply chain and commercial teams early warning signals and market share visibility.

Problem Statement

Cipla operates across 80+ markets spanning generics, branded pharmaceuticals, biosimilars, and API manufacturing — running on a heterogeneous mix of SAP ECC, Veeva CRM, LIMS (LabWare), and MES systems. Data remained siloed across commercial, manufacturing, and regulatory functions with no unified analytics layer.

Challenge 1

01

Fragmented ERP & CRM Data

SAP, Veeva CRM, and MES systems had no unified ingestion pipeline — causing inconsistent commercial and manufacturing KPIs across geographies.

Challenge 2

02

Manual Regulatory Reporting

Schedule M, pharmacovigilance, and WHO-GMP reports were manually collated from multiple source systems — causing T+3 delays and frequent reconciliation errors.

Challenge 3

03

No Real-Time Supply Chain Visibility

Overnight batch jobs provided no intraday view of inventory levels, batch release status, or demand-supply gaps — impacting procurement decisions.

Challenge 4

04

No Unified Customer 360

Doctor outreach, prescription data, and patient support program data were siloed — preventing cohesive market intelligence and territory performance analytics.

Solution Overview

Robust & Scalable Architecture

SAP ECC & S/4HANA data ingested via ADF with CDC & incremental delta loads.

Veeva CRM outputs (call logs, prescriptions) streamed via Azure Event Hubs + Databricks.

LIMS and MES feeds (batch records, QC results) processed via ADF pipelines with schema validation.

Raw → Cleansed → Curated ADLS Gen2 zones with Purview lineage & data cataloging.

Application UI — Power BI Premium

- ▶ DirectQuery to Synapse for real-time dashboards
- ▶ Commercial Sales, Supply Chain & Regulatory modules
- ▶ Row-level security via Azure AD for territory/region access

Source Systems

SAP ECC/S4 | Veeva CRM | LIMS | MES | 3rd-party Feeds

AZURE EVENT HUBS + AZURE DATA FACTORY (Batch & Stream Ingestion)

AZURE DATABRICKS (ETL / Feature Engineering / ML Demand Models)

ADLS Gen2 — Raw → Cleansed → Curated

AZURE SYNAPSE ANALYTICS (DWH + Regulatory Data Mart)

POWER BI PREMIUM — Commercial Sales | Supply Chain | Regulatory | Doctor 360

12+ TB Historical Data | 80+ Markets | 500+ Concurrent Users | 15 Source Systems | 200+ Reports

Business Intelligence Domain

Power BI Premium Dashboards — 5 Core Modules

Commercial Sales Analytics	Doctor 360 & CRM Insights	Supply Chain & Inventory	Regulatory Compliance BI	Finance & P&L Analytics
- Territory & brand-wise sales performance - Market share vs. competitor benchmarks - MR call productivity & coverage ratio - Rolling 12-month trend analysis	- Doctor prescription trends by therapy area - Visit frequency & prescription conversion - Segmentation: high / medium / low prescribers - Territory whitespace opportunity map	- Plant-wise batch release & WIP status - Demand-supply gap & stock-out risk alerts - Distributor secondary sales analysis - Days Inventory Outstanding (DIO) trends	- Schedule M audit readiness tracker - CAPA closure rate & deviation trends - WHO-GMP submission timeline tracker - Pharmacovigilance case status dashboard	- Product-level gross margin analysis - A/R aging & collection efficiency - Opex vs. budget variance by BU - Quarter-close variance commentary

Pipeline Architecture & Engineering Pillars

Batch & Streaming Ingestion

ADF CDC pipelines for SAP ECC delta loads; Event Hubs for Veeva CRM streaming; schema validation gates at every zone boundary.

Lakehouse Architecture

3-zone ADLS Gen2 (Raw / Cleansed / Curated) with Delta Lake format; Databricks Auto Loader for incremental processing.

Transformation & ML Readiness

Databricks notebooks for ETL, feature engineering, and demand forecasting models; orchestrated via ADF trigger pipelines.

Data Quality & Governance

Purview automated classification, 200+ DQ rules in Databricks, lineage tracking from SAP to Power BI report level.

Technical Challenges & Solutions

⚡ Challenge:

SAP extraction timeouts on large FI/CO tables

✓ Solution:

Partitioned ADF parallel copy with pagination token; reduced extraction time by 70%.

⚡ Challenge:

LIMS data variability across manufacturing sites

✓ Solution:

Built site-specific schema adapters in Databricks with master mapping tables for unified output.

⚡ Challenge:

Dataflow refresh bottlenecks during month-end peaks

✓ Solution:

Introduced Fabric capacity burst scaling + priority queuing for critical regulatory pipelines.

⚡ Challenge:

Data lineage gaps for audit trails

✓ Solution:

End-to-end Purview lineage from source SAP tables to Power BI visuals with automated impact analysis.

Ganit's Deliverables and the Value Generated for Cipla

Deliverables & Value Generated

Deliverables

Discovery & Requirements: Conducted stakeholder workshops with Commercial, Supply Chain, and Regulatory teams; delivered a comprehensive BRD covering all 5 BI modules and DE pipeline specs.

Data Architecture & Connectivity: Built ADLS Gen2 medallion architecture with CDC ingestion from SAP, Veeva CRM, and LIMS using ADF and Databricks; implemented OAuth-based secure connectors.

Semantic Model & Security: Developed star schema models in Power BI with 100+ DAX measures; implemented Row-Level Security mapped to BU, geography, and territory hierarchies.

Automated Pipelines: Configured ADF + Databricks orchestration with incremental loads, data quality rules, and SLA alerting; automated month-end report refresh via Power Automate.

Governance & Deployment: Delivered Purview data catalog with automated classification, lineage, and a data dictionary covering 500+ business-critical attributes across source systems.

Value Generated

40% Faster

Report Delivery

Automated pipeline refresh replaced manual collation — delivery from 3 days to same-day for commercial KPI reports.

88% Reduction

Regulatory Errors

Automated Schedule M & pharmacovigilance submissions eliminated manual reconciliation and submission rework.

60% Lower

Infrastructure Cost

Consolidating 4 on-prem warehouses to unified Azure platform achieved elastic scaling at significantly lower TCO.

500+ Users

Self-Serve Analytics

Empowered commercial, finance, and supply chain analysts to independently explore data without SQL expertise.

99.9%

Pipeline Uptime

SLA-monitored ADF pipelines with automated alerting and retry logic ensure near-zero data delivery failures.